Pathology Examination

Gangrene & Edema

Part I: Multiple Choice Questions (MCQs)

1. Gangrene is best defined as:

A) Tissue death due to high blood pressure

✓ B) Massive tissue necrosis followed by putrefaction

C) Accumulation of fluid in serous cavities

D) Regeneration of damaged cells

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2. The primary cause of tissue necrosis leading to gangrene is:

A) Excess vitamin intake

✓ B) Ischemia or toxins

C) Increased lymphatic drainage

D) High-protein diet

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3. The black color of gangrenous tissue is due to formation of:

A) Calcium salts

B) Hydrogen gas

✓ C) Iron sulphide formed by reaction of hydrogen sulphide with iron of hemoglobin

D) Carbon dioxide

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4. The pathogenesis of dry gangrene usually involves:

A) Sudden venous obstruction only

✓ B) Gradual arterial obstruction with patent veins and poor collateral circulation

C) Rapid bacterial infection

D) Excessive tissue fluid accumulation

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5. Which feature is characteristic of dry gangrene but NOT wet gangrene?

A) Rapid spread with severe toxemia

B) Occurs in internal organs

✓ C) Presence of a clear line of demarcation

D) Sudden arterial and venous occlusion

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6. Gas gangrene is caused by contamination of deep wounds with soil containing:

A) Saprophytic organisms

✓ B) Anaerobic bacteria (Clostridium species)

C) Aerobic viruses

D) High levels of Vitamin C

Part II: Essay Questions

Q1. Discuss the Causes of Edema

Definition:

Edema is the **abnormal accumulation of fluid** in interstitial tissues and serous membranes.

A) Localized Causes of Edema

- **Venous Obstruction:**

  Thrombosis, embolism, or external compression (e.g. gravid uterus)

- **Lymphatic Obstruction:**

    - Congenital hypoplasia
    - Inflammation (e.g. filariasis)
    - Fibrosis (post-irradiation)
    - Tumor emboli
    - Surgical removal of lymphatics (e.g. radical mastectomy)
- **Inflammatory Edema:**

  Due to increased capillary permeability in infection or allergy

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B) Generalized Causes of Edema

- **Cardiac Causes (Right-sided heart failure):**
    - Increased capillary hydrostatic pressure
    - Sodium and water retention (aldosterone)
    - Increased capillary permeability due to hypoxia
- **Renal Causes:**

  Proteinuria → ↓ plasma osmotic pressure
- **Nutritional Causes:**

  Protein deficiency or liver disease affecting protein synthesis

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Q2. Compare the Fate of Dry and Wet Gangrene

Dry Gangrene:

- Slow process
- Veins remain patent → fluids drained

- Tissue becomes **dry, shrunken, and mummified**
- **Line of demarcation present**
- May undergo self-separation

Wet Gangrene:

- Sudden arterial and venous occlusion
- No fluid drainage
- Rapid bacterial growth
- **Rapid spread with no line of demarcation**
- Severe **toxemia** and life-threatening